

We return to the setting of vector curves. We've seen that the terminal points of the vector-function $\mathbf{r}(t) = \langle x, y \rangle = \langle g(t), h(t) \rangle$ trace out a smooth path C through the 2-space, where, say, $a \leq t \leq b$. If $w = f(\mathbf{r}(t)) = f(g(t), h(t))$ is some scalar function evaluated along C , then we can draw the following picture:

Our goal is to find the 'area under the curve' of f over the curve C . The natural way to do this is to integrate with respect to *arclength*. We've discussed this notion before—we previously noted:

$$s(t) = \int_a^t \|\mathbf{v}(\tau)\| d\tau$$

With this differential (a.k.a., the *arclength Jacobian*), we are ready to find the desired area:

How to Evaluate a Line (Path) Integral: To integrate a continuous function $f(x, y)$ over an curve C in the xy -plane,

1. Find a smooth parametrization of C :

$$\mathbf{r}(t) = \langle g(t), h(t) \rangle, \quad \text{where} \quad a \leq t \leq b.$$

2. Evaluate the integral as

$$\int_C f(x, y) ds = \int_a^b f(g(t), h(t)) \|\mathbf{v}(t)\| dt.$$

Of course, this can be naturally extended to curves C defined in 3-space as well, where we want to integrate a continuous 4-dimensional surface $w = f(x, y, z)$ over a curve C parametrized by $\mathbf{r}(t) = \langle g(t), h(t), \ell(t) \rangle$ with $a \leq t \leq b$. Since we cannot visualize four dimensions, we settle for drawing C and letting the line integral handle the surface.

Example. Integrate $f(x, y, z) = x - 3y^2 + z$ over the line segment C joining the origin to the point $Q(1, 1, 1)$.

Of course, if our curve C is actually the union of *two* smooth contours (i.e., $C = C_1 \cup C_2$), we can just integrate over each curve separately and add them together.

Example. Consider $O(0, 0, 0)$, $P(1, 1, 0)$, and $Q(1, 1, 1)$. Integrate $f(x, y, z) = x - 3y^2 + z$ over $C_1 \cup C_2$, where $C_1 = \overline{OP}$ and $C_2 = \overline{PQ}$.

Key observation: The value of the integral over C and $C_1 \cup C_2$ are *different*. Thus, *how* we get from the origin to the point $(1, 1, 1)$ seems to matter.

Example. Find the line integral of $f(x, y, z) = 2xy + \sqrt{z}$ over the helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$, where $0 \leq t \leq \pi$.

Finally, we remark that if our curve C turns out to be a *closed* curve—that is, it begins and ends at the same point, then we call it a **contour** and denote its corresponding contour integral by $\oint$.

Example. Find the line integral of $f(x, y) = 6x + 3\sqrt{y}$ over the closed curve $C = C_1 \cup C_2$, where C_1 connects $(0, 0)$ to $(1, 1)$ via the curve $y = x^2$ and C_2 connects $(1, 1)$ to $(0, 0)$ via $y = x$.